VASANTHA DEVI KANDASAMY

Forward Deployed Engineer · AI Engineer · Full-Stack Developer (AI Applied)
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SUMMARY

Master of Computer Applications with 3+ years of web development (React/Node) plus recent data-analytics and applied-AI training. Recently shipped four public, portfolio-grade AI applications: a natural-language-to-SQL analytics copilot, an agentic operations copilot, an AI service advisor for a real Calgary business, and a streaming support assistant. Each is containerized, tested with evaluation suites, and deployment-ready. Comfortable owning a problem from the first customer conversation to a deployed, observable solution.

TECHNICAL SKILLS

AI & LLM Engineering: Claude (Anthropic) API, LLM agents & tool use, RAG, semantic search (pgvector), MCP / FastMCP, prompt engineering, evaluation suites, LLM observability (Langfuse), token streaming (SSE)
Backend & Data: Python (FastAPI), Node.js (Express), SQL, PostgreSQL, MongoDB, MySQL, REST APIs, JWT/OAuth; Pandas, NumPy, data wrangling, visualization & dashboards, statistical analysis
Frontend: JavaScript, React, Redux, Tailwind CSS, Material UI, HTML/CSS/SASS
DevOps & Cloud: Docker, Kubernetes (GKE), Google Cloud (Cloud Run, Cloud SQL), CI/CD, Git
Testing & Practices: Playwright, Cypress, Jest, eval-driven testing, Agile/Scrum, JIRA, Pivotal Tracker

AI PROJECTS

Analytics Copilot · **Natural-language-to-SQL over a database**
Python · FastAPI · Claude API · PostgreSQL + pgvector · React · Docker/GCP · github.com/vasanthadevi/analytics-copilot

- Answers plain-English business questions: retrieves relevant tables via semantic search (RAG with pgvector), generates read-only SQL, self-corrects on errors, and returns a chart, a table, and the exact SQL it ran.
- Enforced a dedicated read-only database role (SELECT-only, statement timeout); authored a 40-question execution-accuracy eval suite and raised accuracy from 97.5% to 100% with targeted prompt changes.

Nimbus Operations Copilot · **Agentic operations assistant with human-in-the-loop approval**
Python · FastMCP (MCP) · Claude API · FastAPI · React · PostgreSQL · Langfuse · Docker/GCP · github.com/vasanthadevi/ops-copilot

- Multi-step agent that plans across tools exposed through a custom MCP server, searching docs, querying customer and billing data, issuing refunds, and opening tickets.
- Enforced human-in-the-loop approval before any write (delivered over a live SSE stream) with full Langfuse tracing and a 10-task eval suite: 10/10 correctness, zero unapproved writes.

Singh's Auto Lab AI Service Advisor · **Agentic web assistant for a real Calgary business**
Python · FastAPI · 3 FastMCP servers · Claude API · React · Google Calendar · Twilio · Docker · github.com/vasanthadevi/singhs-auto-lab-advisor

- Triages a customer's problem and returns a grounded estimate from a structured pricing knowledge base (priced by the data, not the model); books into a real Google Calendar and recognizes returning customers via a phone-keyed CRM, with SMS through Twilio.
- Capabilities split into repointable MCP servers, plus vision photo-diagnosis and an owner dashboard.

AI Support Assistant · **Streaming, grounded customer-support chat**
Node.js · Express · Claude API · PostgreSQL · React · Docker · github.com/vasanthadevi/ai-support-assistant

- Real-time assistant that streams answers token-by-token (SSE) and calls tools to search a knowledge base and look up live order data mid-conversation, citing its sources and refusing to guess.
- Implemented an agentic tool-use loop with grounded responses and per-message token-cost tracking.

EXPERIENCE

Full-Stack / AI Engineer (Volunteer) **CMDSpace, Calgary** — Jul 2026 – Present

- Part of the AI development team for Focus, a task assistant: contributed to a chatbot service (Python, LangChain, RAG over a local LLM) for natural-language task creation, filtering, and queries.
- Worked across the React 19 / TypeScript front end (command-bar Cmd-K UI, Zustand, React Query, Vitest + Playwright) and the FastAPI + SQLAlchemy + PostgreSQL back end, containerized with Docker.

Front-End Developer (Volunteer) **CMDSpace, Calgary** — Jan – Jun 2024

- Built the CMDSpace marketing website (cmdspace.io) to the provided requirements: a responsive React single-page site presenting the company's AI solutions, service sections (ML, neural networks, NLP), and a schedule-a-demo call to action across desktop and mobile.

Freelance Web Developer **Upwork (Haydes Technologies), Remote** — 2013 – 2015

- Developed, maintained, and supported web applications; built secure login/authentication and delivered front-end training to associates.

Software Developer **Mascot eSoft / ISG Novasoft, India** — 2006 – 2007

- Built web applications (SQL Server, VB.NET/ASP.NET, JavaScript/jQuery), including an RFID-based access-control and reporting system.

EDUCATION & CERTIFICATIONS

Master of Computer Applications (MCA), Bharathiar University, India (2006) · **Bachelor of Computer Applications (BCA)**, Bharathiar University, India (2003)
IBM Data Analyst (Coursera, 2024) · NPower Canada Junior Data Analyst (2024) · Microsoft Azure AI-900 (2024)